

Vector Databases in Practice

Why a vector database

A vector database stores embeddings together with their metadata and answers nearest-neighbour queries quickly. Comparing a query against every stored vector becomes slow as a collection grows, so these systems build an index.

A common index is HNSW, a layered graph that lets a search jump across the collection in large steps first and refine its answer in smaller steps, checking only a small fraction of the vectors.

Payloads and filtering

Each stored point carries a payload: a small JSON document with fields such as the source file, the page and the original text. The payload is returned with the search results and can also be used to filter them, for example to search only within one document.

Idempotent ingestion

If the identifier of each point is derived from the content itself, ingesting the same document twice writes the same identifiers again. An upsert operation then replaces the existing points instead of adding duplicates, which makes retries after a failure safe.

Limits and concurrency

Ingestion services usually cap file sizes and the amount of text per request, and limit how many uploads are processed at the same time, so that one large request cannot exhaust memory or slow down search for everyone else.